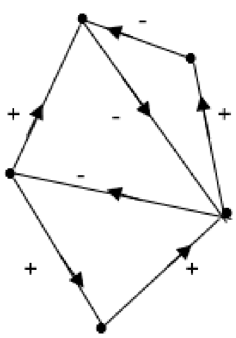
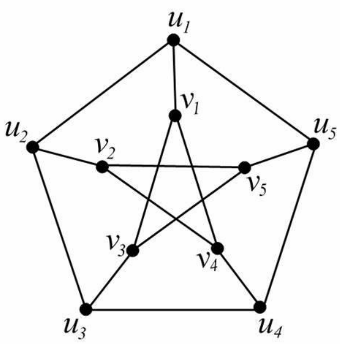
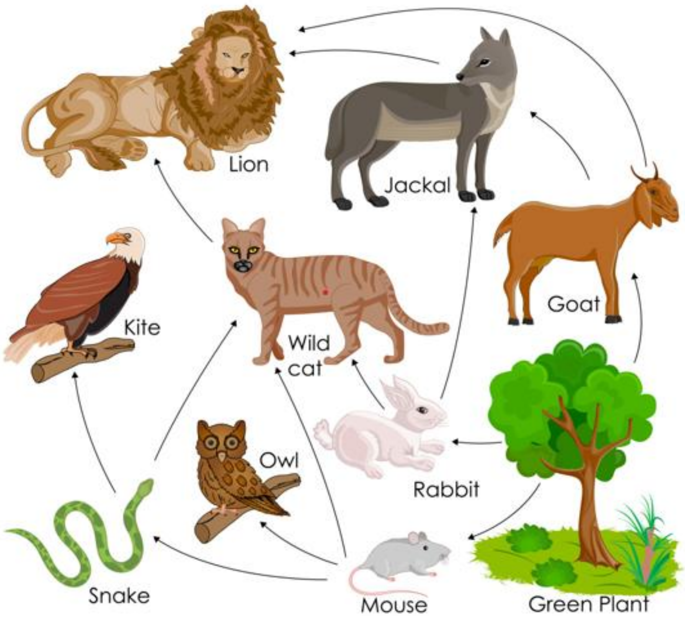
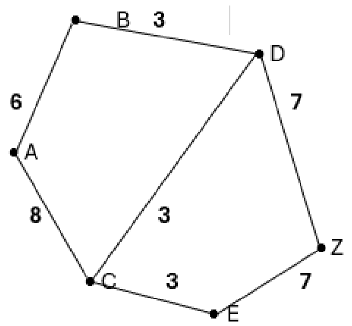
RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Date: 25/06/2025	**Improvement CIE**	**Max. Marks:100**
Semester: VI	**UG**	**Duration: 2 Hrs. (2pm-4pm)**
Course Title: Mathematical Modelling		**Course Code: MA266TEU**

Department of Mathematics

	Quiz			
SL. No	**Questions**	**M**	**BTL**	**CO**
1	Is the given graph G balanced? Find the degree of balance of G: G: 	**2**	**2**	**2**
2	Draw the complete graph $k_{3,4}$	**2**	**1**	**1**
3	Model a real life scenario using a digraph.	**2**	**1**	**1**
4	Define bipartite graph with an example.	**2**	**3**	**2**
5	The chromatic number of the following graph H. 	**2**	**3**	**3**

SL. No	Test	M	BTL	CO
1.	Define the Harary measure of status. Find the Harary measure of each individual in the organizational graph shown in the figure below. 	10	1	1
2.	The Mathematics department of a certain college plans to schedule the classes Graph Theory (GT), Statistics (S), Linear Algebra (LA), Advanced Calculus(AC), Geometry (G) and Modern Algebra (MA) this summer. Ten students given below have indicated the course they plan to take. With this information, use graph theory to determine the minimum number of time periods needed to offer these courses so that every two classes having student in common are taught at different time periods during the day. Of course, two classes having no students in common can be taught during the same period. Anden: LA,S, Brynn: MA,LA,G, Chase: MA, G,LA, Denise: G, LA, AC Everett: AC, LA, S, Francois: G,AC, Greg: GT, MA, LA Harper: LA, GT,S Irene: AC, S, LA Jennie: GT, S	10	2	3

3.	At the end of the school day, all the students plan on driving from school (Vertex A) to the concert (Vertex Z). The edges in the graph below represent one-way roads, and the weights of each edge represents the number of vehicles (in hundreds) that particular road can handle in one hour. What is the greatest number of vehicles that can get from school. 	10	3	3
4.	Evaluate $\int_1^2 \{e^x + (y + xy') \cos xy\} dx$ where y is a solution of the corresponding Euler-Lagrange equation satisfying $y(1) = 2\pi,\ y(2) = \frac{\pi}{2}$.	10	2	2
5.	Find the curve joining two given points A and B so that a particle moving under gravity, along this curve, starting from A reaches B in the shortest possible time, friction and resistance of the medium being neglected.	10	3	4

***** ALL THE BEST *****

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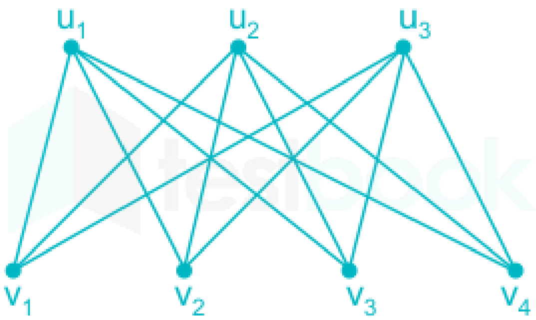
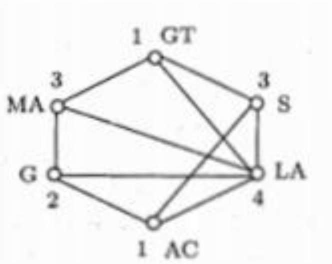
SCHEME & SOLUTIONS

S No	**Solutions with Scheme**	**Marks**
	Part-A	
1.	Unbalanced, degree of balance =2/3	1+1
2.		1+1
3.	bookwork	1+1
4.	bookwork	1+1
5.	$\chi(H) = 3$	2

Q.No	**Part-B**	**Marks**
1	h(Mouse) -1x3+2x2=7, h(snake)= 1x2+2x1=4, h(owl)=0, h(kite)=0, h(wild cat)=1, h(rabbit)=4, h(green plant)=16, h(goat)=1x2=2, h(Jackel)=1x1=1, h(lion)=o	**2+2+ 2+2+2**
2	First, we construct a graph H whose vertices are the six sub-	**1+2 +2+2 +1+2**

	jects. Two vertices (subjects) are joined by an edge if some student is taking classes in these two subjects(see Figure). The minimum number of time periods is $\chi(H) = 4$. 	
3	$c_1 = \{AB, AC\} = 8 + 8 = 14, c_2 = \{BD, DC, CE\} = 3 + 3 + 3 = 9$, $c_3 = \{DZ, EZ\} = 7 + 7 = 14, c_4 = \{AC, BD\} = 8 + 3 = 11$ **Minimum capacity = 9. So maximum vehicles that can get from school are 9.**	2+2+2+2+2
4	$\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) = 0,$ $\cos xy - (y + xy')x \sin xy - \cos xy + (y + xy')x\sin xy = 0$ **Euler Lagrange equation is satisfied by any $y(x)$.** **Any $y(x)$ satisfying necessary differentiable property and $y(1) = 2\pi, y(2) = \frac{\pi}{2}$ is extremal.** $\int_1^2 \{e^x + (y + xy')\cos xy\}dx = \int_1^2 e^x + \int_1^2 \cos xy\, d(xy)$ $= (e^x)_1^2 + [\sin xy]_{xy=2\pi}^{xy=\pi} = e^2 - e$	2 +2+2 4
5	Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ be two fixed point of the hanging cable. Let x- axis be the line of reference. Let ds be the element of arc length of the cable. Let ρ be the density of the cable so that ρds is the mass of the element. If g is the acceleration due to gravity then the potential energy is given by $PE = mgh = (\rho ds)gy.$ Therefore total PE of the cable is given by, $I = \int_P^Q \rho dsgy = \int_{x_1}^{x_2} \rho gy \frac{ds}{dx} dx.$ But $\frac{as}{dx} = \sqrt{1 + (y')^2}$. Therefore $I = \int_{x_1}^{x_2} \rho gy\sqrt{1 + (y')^2}\, dx$ (1) To find the shape of the curve, it is required to extremize the functional given by (1). Consider $f(x, y, y') = \rho gy\sqrt{1 + (y')^2}$. As the functional is independent of x, the Euler equation is given by $f - y' \frac{\partial f}{\partial y'} = c,$ $\Rightarrow \rho gy\left[\sqrt{1 + (y')^2} - y'\rho gy \frac{2y'}{2\sqrt{1+(y')^2}}\right] = c,$ $\Rightarrow \rho g[y(1 + (y')^2) - y(y')^2] = c\sqrt{1 + (y')^2},$	2+2+2 2+2

	$\Rightarrow \rho g y = c\sqrt{1+(y')^2}$, $\Rightarrow \left(\frac{\rho g}{c}\right)^2 y^2 = 1+(y')^2$, $\Rightarrow k^2y^2 = 1+(y')^2 \Rightarrow (y')^2 = k^2y^2-1$, $\Rightarrow y' = \frac{dy}{dx} - \sqrt{k^2y^2-1}$, $\Rightarrow \frac{dy}{\sqrt{k^2y^2-1}} = dx$, $\frac{dy}{k\sqrt{y^2-\frac{1}{k^2}}} = dx$. Integrating, $\cosh^{-1}\left(\frac{y}{1/k}\right) = kx + k_1$, $y = \frac{1}{k}\cosh(kx+k_1)$, which represents a catenary. Therefore a cable between two fixed points hangs in the form of a catenary.	

SCRUTINY & EVALUATION OF CIE QUESTION PAPER

Date: 02/05/2025	**Test - 1**	**Max. Marks:100**
Semester: VI	**UG**	**Duration: 2 Hrs. (2pm-4pm)**
Course Title: Mathematical Modelling		**Course Code: MA266TEU**

Declaration by the Course handling faculties:

As a teaching faculty of the course: **Mathematical Modelling** with code: **MA266TEU**

We hereby confirm that the question paper with Scheme and Solutions is thoroughly reviewed and we ensure that it adheres to the following criteria:

Sl. No	Criteria	Yes/No
1	The question paper adequately covers the prescribed syllabus contents.	
2	The question paper is in line with the recommended pattern, taking into consideration the structure and format suitable for the Continuous Internal Evaluation.	
3	The question paper is designed to align with the Revised Bloom's Taxonomy, encompassing various levels of cognitive skills such as remembering, understanding, applying, analysing, evaluating, and creating.	
4	The question paper is aligned with the defined course outcomes, ensuring that it effectively assesses the knowledge and skills acquired during the course.	
5	Course handling faculty (as applicable) who are responsible for preparing the question paper, Scheme and Solutions have unanimously agreed to utilize this Question paper for conducting the Continuous Internal Evaluation.	
6	The Question paper, Scheme and complete Solutions have been submitted to the Test	

	coordinators within the designated time-frame to ensure the smooth conduction of Continuous Internal Evaluations	

Course handling Faculties:

Name: **Signature**

1. Dr. Y. Sailaja

2. Dr. Nivya Muchikel

To be filled by the Scrutinizer:

Sl. No	Rubrics	Points	
		Max	**Awarded**
1	Timely submission of the question paper along with the scheme & solution	10	
2	Heterogeneous nature of QP with respect to BTs and Cos	10	
3	Format with proper entry of all particulars including test, course name, code, date, max marks, BT CO table, efficient use of paper (proper spacing, figures)	10	
4	No handwritten data or diagrams, and uniform fonts throughout	10	
5	Scheme & complete Solutions in the format	10	
	Total points	**50**	
Any other comments by the scrutinizer :			

Note: Course coordinators to obtain scrutinizer's acceptance by incorporating all suggestions from scrutiny into the final versions of QP, Scheme, and Solutions.

All corrections suggested by the scrutinizer are incorporated and both the copies are re-submitted	**Signature of Course coordinator**

Accepted/Rejected	**Signature of Scrutinizer** **(Name:)**

Signature of HOD